LahjaSensei at DialectSentEval 2026: Few-Shot Prompting and Model Merging for Sentiment Swap

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Abstract

Arabic dialect sentiment swapping requires systems to reverse the affective orientation of an utterance while preserving its core meaning, dialectal form, and fluency. This task is particularly challenging because sentiment may be conveyed through dialect-specific vocabulary, negation patterns, and spelling variation. We present LahjaSensei, our system for Subtask 2 of the DialectSentEval shared task. We investigate two complementary approaches: prompt-based generation with GPT-5.6 and parameter-efficient fine-tuning of Qwen2.5-7B. We evaluate GPT-5.6 under zero-shot, retrieved few-shot, stronger Arabic instructions, and direction-aware configurations. The direction-aware GPT-5.6 pipeline explicitly predicts source polarity before retrieving direction-matched demonstrations and generating the rewrite. For Qwen2.5-7B, we compare pooled QLoRA with linear and TIES merges of direction-specific adapters. Direction-aware GPT-5.6 prompting achieves the highest test sentiment accuracy and remains stable across development and test. Pooled QLoRA produces the highest development BLEU and chrF, while linear merging obtains the highest test BLEU and chrF but lower sentiment accuracy. These findings highlight the central trade-off between sentiment control and preservation of semantic content and demonstrate the value of explicit direction modeling for Arabic dialect sentiment swapping.

1 Introduction

Sentiment style transfer modifies a text's affective orientation while preserving its semantic content. Unlike sentiment classification, sentiment swapping requires generating an output with the target polarity while retaining the source's events, entities, and central message. These goals can conflict: conservative rewrites may leave sentiment cues unchanged, while aggressive rewrites may cause semantic drift.

This challenge is particularly acute in dialectal Arabic. Sentiment may be expressed through dialect-specific vocabulary, morphology, negation patterns, orthographic variation, code-switching, hashtags, and emoji. DialectSentEval 2026 extends prior work on Arabic sentiment analysis and swapping (Ezzini et al., 2026b) across Maghreb, Egyptian, Gulf, and Levantine dialect groups. In Subtask 2, systems must rewrite a dialectal Arabic sentence to express the opposite sentiment while retaining its core meaning (Ezzini et al., 2026a). The task therefore jointly evaluates dialect understanding, sentiment control, and preservation of the source content during generation.

LahjaSensei investigates two complementary modeling approaches. The first uses GPT-5.6 (OpenAI, 2026) in a direction-aware pipeline for sentiment swapping that first predicts the source polarity and then retrieves demonstrations from the corresponding transformation direction before generating the rewrite. The second adapts Qwen2.5-7B (Qwen Team, 2024) using QLoRA (Dettmers et al., 2023). We compare a pooled adapter trained jointly on both directions with linear merging and TIES-Merging (Yadav et al., 2023) of separately trained directional adapters.

Our experiments show that direction control produces the most reliable sentiment inversion. Pooled QLoRA achieves the best development BLEU and chrF, while linear merging obtains the best test BLEU and chrF but lower sentiment accuracy. These findings highlight the trade-off between polarity control and preservation of content.

Our main contributions are:

- We develop a direction-aware few-shot pipeline combining source-polarity prediction with direction-matched retrieval.
- We compare zero-shot and few-shot prompting with pooled QLoRA and linear and TIES merges of direction-specific adapters.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1403–1408
October 28-29, 2026 ©2026 Association for Computational Linguistics

- We analyze the trade-off between polarity inversion and content preservation, including the contribution of each pipeline stage to sentiment failures.

Code and prompts are available at our repository.[1]

2 Background

2.1 Task and Dataset

Task. DialectSentEval Subtask 2 focuses on sentiment swapping in Arabic dialects. Training and development instances include source polarity, and systems must generate the opposite polarity while preserving the source's entities, events, and meaning (Ezzini et al., 2026a). For example, "a very wonderful program" is rewritten as "a very bad program." The released test file contains instance identifiers and source sentences; therefore, systems must infer the transformation direction at inference time. LahjaSensei participated only in Subtask 2.

Dataset. We use MA'AKS Extended, the DialectSentEval shared-task dataset, which builds on the manually curated MA'AKS parallel corpus (Mughaus et al., 2026). It contains 6,752 training, 964 development, and 1,931 test instances spanning Maghreb, Egyptian, Gulf, and Levantine dialect groups. Training and development instances include source polarity and an opposite-polarity reference rewrite, while the released files do not provide dialect labels for individual instances.

2.2 Related Work

Arabic sentiment swapping can be viewed as a controlled text generation task in which sentiment is modified while semantic content is preserved. MA'AKS provides manually curated positive–negative sentence pairs for studying this problem (Mughaus et al., 2026). Its annotation analysis shows that human rewrites rely primarily on lexical substitution and negation, while many examples require combinations of multiple editing operations. These findings demonstrate that sentiment inversion cannot be reduced to simply adding or removing a negation marker. The MA'AKS experiments also show the potential of both prompting and parameter-efficient fine-tuning for Arabic sentiment transfer.

Recent Arabic sentiment-swapping systems have explored decomposed prompting and supervised generation. Alami et al. (2026) use separate LLM calls to identify sentiment-bearing expressions, generate their opposites, and reconstruct the sentence. This decomposition makes intermediate steps explicit but may propagate upstream errors and introduce unnecessary changes. Bichi and Yassine (2026) fine-tune mT5-small with an Arabic target-polarity instruction, offering a simpler end-to-end formulation but frequently copying the source without sufficiently reversing its sentiment. Prompt design is also important in dialectal Arabic, where the benefit of aligning the instruction language or dialect with the input varies across models (Alharbi et al., 2025).

These findings motivate our comparison of Arabic prompt formulations, retrieved demonstrations, explicit direction control, and merged direction-specific Qwen adapters. Unlike the decomposed pipeline of Alami et al. (2026), our main method uses one polarity-prediction call followed by one direction-conditioned whole-sentence rewrite. This design reduces intermediate generation stages, which may limit opportunities for error propagation and unnecessary content changes while retaining explicit control over the transformation direction.

3 System Overview

3.1 GPT-5.6 Prompting

All GPT-based systems use the `GPT-5.6` model through the OpenAI Responses API, with reasoning effort set to `none`, text verbosity to `low`, and temperature and top-p left at their default values.

Two-stage directional prompting. Our submitted GPT-5.6 system uses a two-stage direction-aware pipeline: it first predicts the source polarity, then rewrites the input in the opposite direction using four direction-matched demonstrations. The pipeline is defined as

$$\begin{aligned} \hat{s} &= f_{\text{cls}}(x), \\ D_x &= \text{Retrieve}_4(x \mid s = \hat{s}), \\ y &= f_{\text{rw}}\big(x, \hat{s}, \text{opp}(\hat{s}), D_x\big). \end{aligned}$$

Here, x denotes the source sentence, $\hat{s}$ its predicted polarity, and D_x the retrieved demonstration set. The rewriting prompt explicitly specifies the source and target polarities and requests a minimal, fluent rewrite while preserving the dialect, entities, events, numbers, punctuation, and non-sentiment emoji.

[1] https://github.com/ahmadalmoustafa/LahjaSensei-DialectSentEval-2026

Single-stage prompt variants. For comparison, we evaluate four single-stage baselines using two Arabic prompt formulations under zero-shot and retrieved four-shot settings. The original prompt requests a minimal rewrite expressing the opposite sentiment. The stronger prompt adds preservation constraints for the dialect, entities, events, numbers, punctuation, and non-sentiment emoji while changing only sentiment-bearing expressions. Unlike the two-stage system, these baselines directly request sentiment reversal without predicting and specifying the source polarity. In the four-shot settings, retrieved source–target pairs are presented as separate user–assistant exchanges.

Demonstration retrieval. The demonstration-selection is based on the same procedure, with the two-stage system additionally restricting candidates to the predicted transformation direction. Demonstrations are drawn exclusively from the training data. Arabic text is normalized by removing diacritics and tatweel and standardizing common orthographic variants. Candidate pairs are filtered to remove unchanged pairs, exact source matches, and examples that fail preservation, similarity, or length constraints.

Relevance combines character- and word-level TF–IDF similarity. We then apply maximum marginal relevance (MMR) (Carbonell and Goldstein, 1998) to select four demonstrations that balance relevance and diversity. For the single-stage baselines, candidates may represent either transformation direction. For the two-stage system, retrieval is restricted to examples whose source polarity matches $\hat{s}$, so all four demonstrations represent the same positive-to-negative or negative-to-positive direction.

3.2 Qwen2.5-7B Adapter Merging

We evaluate three configurations based on `Qwen/Qwen2.5-7B-Instruct`. The pooled baseline fine-tunes one QLoRA adapter on both transformation directions. Its prompt specifies the source and target polarities and requests only the rewritten Arabic sentence. The other configurations fine-tune separate positive-to-negative and negative-to-positive adapters.

We combine the direction-specific adapters through weighted adapter merging. Linear merging averages their parameter updates with equal weights. TIES-Merging removes low-magnitude updates, resolves sign conflicts, and combines updates that agree in direction. We use unit adapter weights and a TIES density of $d = 0.2$. At inference time, CAMeLBERT (Inoue et al., 2021) determines the source polarity by comparing its positive and negative probabilities, and the opposite label specifies the target generation direction.

4 Experimental Setup

We use the official training, development, and test partitions. The development set guides prompt selection, retrieval design, and Qwen configuration selection. Test labels and reference rewrites remain hidden during development, and the shared-task organizers compute the official test scores.

We evaluate five GPT configurations: four single-stage systems combining the original and stronger Arabic prompts with zero-shot and retrieved four-shot settings, and the two-stage directional system. For Qwen, we compare the pooled QLoRA baseline with linear and TIES merges of direction-specific adapters. The development search considers four candidate linear weighting schemes and three alternative TIES densities, from which we report the best configuration of each merging method. Each configuration is evaluated once; therefore, run-to-run variation is not estimated. Complete implementation settings are provided in Appendix C.

The official evaluation uses sentiment accuracy, BLEU (Papineni et al., 2002), and chrF (Popović, 2015). Sentiment accuracy is computed with the dialectal CAMeLBERT classifier; BLEU and chrF assess preservation of semantic content through overlap with the reference rewrite.

5 Results and Analysis

5.1 Quantitative Results

Tables 1 and 2 compare performance on the development and test sets. Higher BLEU and chrF do not consistently correspond to higher sentiment accuracy, highlighting the tension between preserving content and reversing polarity.

The two-stage directional system achieves the highest test sentiment accuracy of 0.779, compared with 0.693 for the original four-shot system. This improvement comes with a decrease in BLEU from 44.76 to 41.48 and in chrF from 65.95 to 64.45. The directional system also remains stable across splits, changing from 0.785 on development to 0.779 on test. In contrast, the original four-shot

System	BLEU	chrF	Sent. Acc.
GPT directional 4-shot	42.84	64.01	0.785
GPT stronger 4-shot	43.16	63.98	0.717
GPT original 4-shot	46.81	66.38	**0.809**
GPT stronger zero-shot	42.22	63.45	0.682
GPT original zero-shot	46.01	65.97	0.771
Qwen pooled QLoRA	**53.08**	**71.21**	0.784
Qwen linear merge	51.72	68.91	0.689
Qwen TIES merge	50.09	67.92	0.711

Table 1: Development results. The best score for each metric is shown in bold.

System	BLEU	chrF	Sent. Acc.
GPT directional 4-shot	41.48	64.45	**0.779**
GPT stronger 4-shot	42.26	64.54	0.694
GPT original 4-shot	44.76	65.95	0.693
GPT stronger zero-shot	40.56	63.52	0.696
GPT original zero-shot	43.40	65.28	0.662
Qwen pooled QLoRA	49.82	68.11	0.652
Qwen linear merge	**49.90**	**68.56**	0.627
Qwen TIES merge	47.72	67.13	0.608

Table 2: Official test results. The best score for each metric is shown in bold.

system decreases from 0.809 to 0.693. Explicit direction prediction therefore provides more consistent sentiment control.

Retrieved demonstrations improve test BLEU and chrF for both prompt formulations relative to their zero-shot variants. Their effect on sentiment accuracy is less consistent. Four-shot prompting improves the original prompt but slightly reduces the accuracy of the stronger prompt. Retrieved examples therefore improve BLEU and chrF but do not guarantee polarity reversal without explicit direction control.

Among the Qwen configurations, pooled QLoRA achieves the highest development scores, with 53.08 BLEU, 71.21 chrF, and 0.784 sentiment accuracy. On the test set, linear merging slightly improves BLEU from 49.82 to 49.90 and chrF from 68.11 to 68.56 compared with pooled QLoRA. However, its sentiment accuracy decreases from 0.652 to 0.627. TIES underperforms pooled QLoRA on all three test metrics. Thus, merging direction-specific adapters offers no consistent advantage over joint bidirectional fine-tuning, reinforcing the need to interpret all three metrics jointly.

5.2 Error Analysis

We analyze the two stages of the directional GPT-5.6 system to identify where sentiment failures arise. On the development set, GPT-5.6 correctly predicts the source polarity for 880 of 964 examples, while CAMeLBERT assigns the target polarity to 757 final outputs. Since the number of correct direction predictions exceeds the number of successful final outputs, at least 123 examples with correctly predicted source polarity still fail to express the target sentiment after rewriting. This indicates that direction prediction is not the sole source of error and that failures also arise during the rewriting stage.

Manual inspection of the test outputs identifies three content-related error types. Incomplete cue revision can produce internally inconsistent sentences, semantic drift may change the state of an event, and numerical alteration may modify factual details during polarity reversal. Representative examples are presented in Appendix A.

6 Conclusion

We presented LahjaSensei, our system for DialectSentEval 2026 Subtask 2. We compared GPT prompting, pooled QLoRA, and linear and TIES merges of direction-specific Qwen adapters. Direction-aware prompting achieved the most reliable inversion, while pooled QLoRA produced the highest development BLEU and chrF and linear merging the highest test BLEU and chrF. Merging did not consistently outperform pooled fine-tuning, highlighting the trade-off between polarity control and preservation of semantic content. Analysis shows that correct direction prediction does not guarantee successful rewriting. Future work includes repeated experiments under additional controlled settings, broader error analysis, and human evaluation.

7 Limitations

Our study has several important limitations. Each configuration is evaluated only once during our experiments, without human evaluation of semantic preservation, fluency, or dialectal naturalness. Missing instance-level dialect labels prevent detailed dialect-specific analysis, and the manual error categories remain subjective.

Acknowledgments

We thank the DialectSentEval 2026 organizers for coordinating the task, providing data, and maintaining the evaluation platform. This work was supported by JST CREST JPMJCR2565, Japan.

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A Qualitative Examples

Table 3 presents three representative errors from the submitted two-stage directional GPT-5.6 system. The examples illustrate incomplete cue revision, semantic drift, and alteration of numerical details.

ID	Example
3036	**Type:** Incomplete cue revision **Source:** سيئ جدا للأسف يتم التوقف بشكل كبير **Output:** ممتاز جدا لحسن الحظ يتم التوقف بشكل كبير
3011	**Type:** Semantic drift **Source:** للأسف التطبيق لا يعمل بعد التحديث الاخير الرجاء حل المشكله **Output:** لحسن الحظ التطبيق يعمل بعد التحديث الاخير شكرًا لكم على حل المشكله
2378	**Type:** Numerical alteration **Source:** اطلب بيتزا هت خمسه يعطوني 3 اش هذا النصب **Output:** اطلب بيتزا هت خمسه يعطوني 5 اش هذا الكرم

Table 3: Examples of manually identified errors in the submitted two-stage directional GPT-5.6 system.

B Prompt Templates

The submitted system uses separate Arabic prompts for source-polarity prediction and directional sentiment rewriting. Faithful English

translations are provided below, preserving the original instructions and output constraints.

B.1 Source-Polarity Classification

System message.

> Act as a classifier specialized in identifying the overall sentiment of Arabic texts written in different dialects. Classify the sentence as positive or negative. Select the dominant sentiment even when the sentence contains mixed expressions. Base the decision on the speaker's attitude rather than an isolated word. Do not rewrite the sentence or provide an explanation. Return only one label: positive or negative.

User template.

> Determine the overall sentiment of the following sentence:
>
> `{source}`

The returned label is parsed as the predicted source polarity and reversed to obtain the target polarity.

B.2 Directional Rewriting

System message.

> Act as a system specialized in reversing sentiment in Arabic texts written in different dialects. Rewrite the sentence from its current polarity to the requested polarity while preserving its core meaning. Change as few sentiment-bearing words or expressions as possible. Preserve the topic, events, entities, actions, roles, locations, numbers, and other factual details. Maintain the original dialect, writing style, and word order where possible, and do not convert dialectal text into Modern Standard Arabic. Preserve links, mentions, hashtags, and non-sentiment emoji. Make the target sentiment explicit rather than relying on negation alone. Do not introduce new reasons, events, or information. Return only the rewritten sentence without labels, explanations, quotation marks, or alternative outputs.

User template.

> Current sentiment: `{source_polarity}`
> Target sentiment: `{target_polarity}`
> Sentence: `{source}`

B.3 Four-Shot Message Construction

The four retrieved demonstrations are presented as alternating user–assistant messages. All demonstrations represent the same transformation direction as the predicted source polarity:

> **User:** Current sentiment, target sentiment, and retrieved source.
> **Assistant:** Retrieved target rewrite.
> ⋮
> **User:** Current sentiment, target sentiment, and actual input.

The final user message contains the test input, while the preceding four exchanges contain the direction-matched training examples selected by the retrieval method.

C Implementation Details

GPT inference. All GPT experiments use the `gpt-5.6` alias through the Responses API, with reasoning effort set to `none` and text verbosity to `low`. Temperature and top-p are left at their default values. Classification and rewriting use maximum output lengths of 16 and 128 tokens, respectively. API requests use a 120-second timeout and up to five retries.

Retrieval. Arabic text is normalized by removing diacritics and tatweel and unifying common Alef forms and Alef Maqsura. We exclude unchanged pairs, exact source matches, and examples that fail basic preservation, similarity, or length constraints. Relevance combines character- and word-level TF–IDF similarity. From the 40 highest-scoring candidates, maximum marginal relevance (MMR) selects four diverse demonstrations using $\lambda = 0.8$. For the two-stage system, candidates are restricted to the predicted transformation direction.

Qwen training. The pooled adapter and both directional adapters are trained for three epochs with sequence length 512, batch size 1, and gradient accumulation over 16 steps. We use QLoRA with four-bit NF4 quantization, double quantization, LoRA rank 16, $\alpha = 32$, and dropout 0.05. LoRA is applied to the attention and feed-forward projection layers. Optimization uses `paged_adamw_8bit`, a learning rate of 2×10^{-4}, cosine scheduling, and a warmup ratio of 0.05.

Merging and generation. Linear merging uses weights $(0.5, 0.5)$, while TIES uses weights $(1.0, 1.0)$ and density 0.2. Generation is greedy with batch size 8, maximum input length 512, and maximum output length 96.